R22

Code No: 181AA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, September - 2023
APPLIED PHYSICS
(Common to CE, ME, ECE, EIE, AE, BT, MIE, PCE, CSE(AI&ML), CSE(IOT), AI&DS, AI&ML)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A

(10 Marks)

1.a) What is the difference between classical and quantum mechanics? [1]
b) If the momentum of a particle is increased to four times, then find the change in de-Broglie wavelength? [1]
c) What will be the temperature co-efficient of an intrinsic semiconductor? [1]
d) How the bulk resistance of the semiconductor changes as the doping increases? [1]
e) Give two examples of paramagnetic material. [1]
f) What happens to a ferromagnetic material heated above curie temperature? [1]
g) What is Nanoscale? [1]
h) What is the procedure in the top-down synthesis of nanomaterials. [1]
i) Define numerical aperture. [1]
j) Can we obtain light amplification in the absence of stimulated emission? [1]

PART - B

(50 Marks)

2. Derive the expression for time-independent Schrodinger wave equation. [10]

OR

3.a) What are the assumptions in Drude-Lorentz theory?
b) Explain Davisson-Germer experiment in detail? How does it explain the wave nature of electrons? [4+6]

4. Explain with suitable diagrams the conduction band, valence band and the forbidden band. Discuss the contribution of electrons and holes to electrical conduction? [10]

OR

5.a) How n-type and p-type semiconductors are produced?
b) Distinguish between Zener and avalanche breakdown? [5+5]

6.a) Define the following terms (i) Dielectric Polarization, (ii) Polarisability, (iii) Dielectric Constant, (iv) Spontaneous polarization, (v) Electric susceptibility.

b) What is a bubble memory? What type of materials are used in it? [5+5]

OR

7.a) Explain how the dielectric constant of a ferroelectric material varies with temperature. Name two ferroelectric materials?

b) What are the applications of ferroelectric materials? [7+3]

8.a) List the similarities and differences between SEM and TEM.

b) Explain Quantum confinement of nanoparticles. [8+2]

OR

9.a) Explain in detail PVD and CVD methods of nano synthesis of films and relative merits.

b) Write a note on surface to volume ratio. [6+4]

10.a) Differentiate between three level and four level lasers by giving suitable examples.

b) Discuss merits and demerits of single mode optical fibers. [6+4]

OR

11.a) Specify three types of possible energy transitions between two atomic energy levels and also derive conditions for Einstein's coefficients.

b) Light gathering capacity of an optical fibre is 0.479. If relative core cladding index difference is 0.005, calculate the refractive index of cladding if the outside medium is air. [8+2]

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, March/April - 2023
APPLIED PHYSICS
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PART- A

(10 Marks)

1.a) What is blackbody? [1]
b) Define Symmetry in solids [1]
c) State Hall effect. [1]
d) List out applications of BJT. [1]
e) State pyroelectric. [1]
f) What are the applications of Energy Materials? [1]
g) Define Nano. [1]
h) Illustrate applications of nanomaterials. [1]
i) What is acronym LASER? [1]
j) What is total internal reflection? [1]

PART - B

(50 Marks)

2.a) Explain Stefan-Boltzmann's law.
b) Discuss Born interpretation of the wave function. [5+5]

OR

3.a) List out assumptions of Drude & Lorentz free electron theory.
b) Explain Fermi-Dirac distribution of electrons. [5+5]

4.a) Explain working principle of Zener diode.
b) Illustrate working mechanism of PIN diode in forward and reserve bias. [5+5]

OR

5.a) With a neat diagram, describe working principle of Avalanche Photo Diode (APD).
b) Distinguish between intrinsic and extrinsic semiconductors. [5+5]

6.a) What is ferroelectricity? Explain properties of ferroelectric materials.
b) Write a note on bubble memory devices. [5+5]

OR

7.a) Write a note on multiferroics.
b) Explain construction and working principle of rechargeable ion batteries. [5+5]

8.a) Explain quantum confinement phenomenon.
b) Discuss fabrication of nanomaterials using Physical Vapor Deposition (PVD). [5+5]

OR

9.a) Write a note on combustion methods.
b) Discuss surface to volume ratio in nanomaterials. [5+5]

10.a) Describe construction and working mechanism of Nd:YAG laser.
b) Write a note on optical fiber for communication system [5+5]

OR

11.a) Discuss construction and working principle of Argon ion Laser.
b) Derive an expression for acceptance angle numerical aperture. [5+5]

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JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, January/February - 2024
APPLIED PHYSICS
(Common to CE, ME, ECE, EIE, AE, BT, MIE, PCE, CSE(AI&ML), CSE(IOT), AI&DS, AI&ML)

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PART- A

(10 Marks)

1.a) State the Planck's radiation law. [1]
b) Draw E-K diagram of free electron. [1]
c) Why silicon is not used to make LED? [1]
d) Write the difference of Diode and Zener Diode. [1]
e) Distinguish polar and nonpolar dielectrics [1]
f) Write the significance of hysteresis loop in magnetism. [1]
g) Classify the Nanomaterials according to Quantum confinement. [1]
h) Mention the nanoscale for which a material can be called as nanomaterial. [1]
i) Give an example of each atomic laser and molecular laser. [1]
j) Which optical fiber is used for long distance communication? [1]

PART - B

(50 Marks)

2.a) Explain the concept of the Heisenberg's uncertainty principle. Using Heisenberg's uncertainty principle, prove that electron cannot stay in the nucleus of an atom.
b) Describe the findings of photoelectric effect.
c) How does the classical theory fail to explain the photoelectric effect? [5+2+3]

OR

3.a) What does it mean by free and bound electrons? Discuss the important postulates of free electron theory of metals.
b) Explain the Bloch's theorem and show how it leads to energy band stature of solids. [5+5]

4.a) Derive voltage expression and working of the Bipolar Junction Transistor (BJT).
b) Describe construction and principle of LED. [5+5]

OR

5.a) Describe the I-V characteristics of solar cell and photo diode.
b) Write the working principle of PIN and Avalanche photo detectors. [5+5]

6.a) What do you mean by internal field? Derive the expression for internal field for solids.
b) Derive Clausius-Mososotti relationship for cubic solids. [7+3]

OR

7.a) Analyse the working of liquid crystal display.
b) Analyse the working and characteristics of solid fuel cells. [5+5]

8.a) Define bottom-up fabrication method of nano materials.
b) Explain Sol-Gel synthesis for producing nanomaterials with proper diagram of each steps. [3+7]

OR

9.a) How the characterization techniques XRD and SEM are used for nanomaterials? Explain.
b) Write application of nanomaterials for medical technology and civil engineering. [7+3]

10.a) Describe construction and working mechanism of Argon ion Laser.
b) Write a note on optical fiber for communication system. [5+5]

OR

11.a) With neat diagram, explain Nd: YAG laser system.
b) Explain construction of optical fiber. [6+4]

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